Yuan 2018 high-grade glioma — analysis summary

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

8 samples intermix per-neighbourhood (iLISI 1.0→4.08/8) on one continuous manifold (single kNN component, 0 islands); no per-patient archipelago.

Gate 2 — module geography → GO (non-branching)

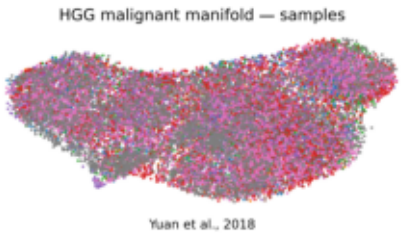
Gast/pSom co-vary (r=+0.23) and both oppose Crest (−0.66 / −0.75): one developmental axis, Crest an inverse terminal gradient. Non-branching case.

Gate 3 — cross-repression → NA

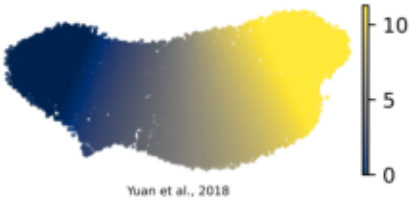
No bifurcation (curve: 2 tips, 0 forks, fold-free across params) → not tested.

STEP FIGURES

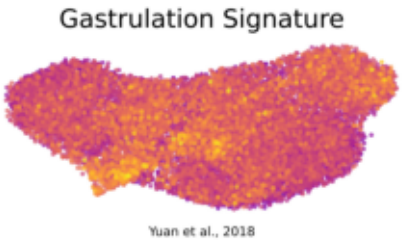
Manifold — samples



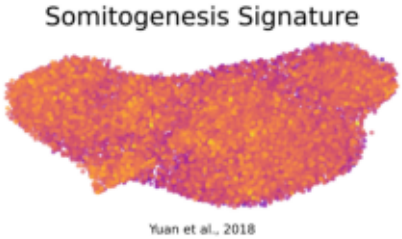
Pseudotime



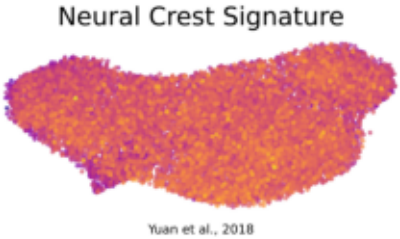
Gastrulation



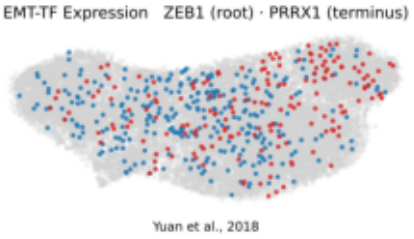
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
High-grade glioma (Yuan et al., 2018), microwell array-based platform (UMI counts) — 20,080 malignant cells (8/8 samples); embryonic-program cell-state analysis.

Verdict.
The embryonic-program pattern reproduces as the non-branching case: a single continuous gastrulation/somitogenesis → neural-crest developmental axis, rooted at the Gast/pSom-high pole with Crest as an inverse terminal gradient. No Gast–pSom bifurcation (Gate 3 NA). Linear-pSom/Crest geometry (pSom μ=+0.035 globally strongest).

Parameter choices

Min-cell threshold	≥100 floor (largest gap 2012→598, r=3.36); 8/8, 20,080 cells	**Genes / embedding**	729 overdispersed (GAM); n_pcs=20; Harmony θ=10
Islands removed	0 (single connected kNN component)	**Geometry**	Non-branching; scFates curve N=20, λ=0.3, μ=100 (2 tips, 0 forks)
Root	curve tip 0 = Gast/pSom-high, Crest-low (Crest is late)	**Module means (raw)**	pSom μ=+0.035 globally strongest (Linear-pSom/Crest)
EMT-TFs	ZEB1 (14%, root) · PRRX1 (9%, terminus); Snail/Twist/FoxC near-absent		